

Digital Storytelling for the Support of Maternal and Child-wellbeing Through Community-based Co-design

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ABSTRACT

In rural, under-resourced, and marginalised regions, digital health information has become a tool with which to reach new and expecting parents for the care of their and their children's wellbeing. However, the types of health information shared is often inaccessible since "top-down" approaches are utilised in creating and sharing content, with limited input by recipients of the information. Furthermore, access to health information has been prioritised over the learning thereof, indicating a need to refocus efforts on how health information is curated. I aim to explore how community-based co-design can be leveraged to design and develop digital storytelling content for the support of maternal and child-wellbeing (MCWB). With the consideration of intersectional feminist methods, participatory visual techniques, and Lambert's taxonomy of participatory media practices, I seek to understand existing storytelling and communication practices, as well as dissemination mechanisms in two rural South African communities for the support of MCWB. This will be achieved through four phases, namely *understanding*, *making*, *connecting*, and *embracing*, using in-depth interviews, photovoice, generative workshops, and focus groups. I am interested in how various community stakeholders participate in the MCWB and health information sharing ecosystem, thus I will be including mothers, fathers, other caregivers, healthcare workers and community leaders. With the creation of digital story artefacts, community-based methods and techniques, and an understanding of health information content creation and sharing via community-based co-design, the intended contributions are: describing how existing communal practices can influence novel approaches to promoting MCWB; developing an understanding of how different degrees of participation in the creation of digital stories impact how people create, share, and learn from digital stories; advances in community-based co-design for rural and under-resourced populations; and explaining how digital stories could impact MCWB within a community.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); HCI design and evaluation methods; Field studies.

KEYWORDS

community-based co-design, digital storytelling, maternal and child-wellbeing,

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1 INTRODUCTION

New and expecting parents are often overwhelmed and stressed with preparing for the birth of their child and consuming new information on caring for their child and themselves. Digital health has provided opportunities to reach parents in under-resourced and marginalised communities; however, non-governmental organisations (NGOs) and governmental health departments are prone to share health information in inaccessible forms [1, 2]. This is due to technology with limited functionalities used by parents and the types of content shared by health authorities. The dominant approach of sharing health information has entailed the creation of health messages by government or NGOs, which have been appraised by experts, i.e. a 'top-down' approach [3]. There is a disconnect between the messaging delivered and lived practices in under-resourced contexts which has led to a failure in meeting maternal and child wellbeing (MCWB) objectives [4, 5]. The receipt of health information by target populations has, in many cases, been prioritised over the *learning* by the targeted populations of what the information means, and how an understanding of information informs action [6]. A consideration of existing cultural practices, as they relate to communicating health information and experiences of new and expecting mothers, can illuminate opportunities to support MCWB in under-resourced communities [7]. Storytelling is such a practice, and I posit that exploring ways to support digital storytelling (DST), i.e. using digital technologies to tell stories, can work towards resolving the disconnect between the dissemination, understanding and acting upon information, and the challenge of communicating effective health information and experiences in under-resourced communities.

2 RESEARCH AIM

This project seeks to explore how community-based MCWB care can be supported through storytelling via digital media, by co-designing digital story material within rural, under-resourced communities. It will do so by understanding barriers to participation in designing storytelling media artefacts through the use of collaborative activities. The project will employ an intersectional feminist

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community-based participatory action research (CBPAR) methodology to collaboratively design digital story media artefacts with community members. By leveraging existing storytelling and embedded communication practices within the community, different levels of engagement in digital media creation, termed participatory media, will be explored with community members, in an effort to support the wellbeing of mothers and children. The main contributions intended by this research are: describing how existing communal practices can influence novel approaches to promoting MCWB; developing an understanding of how different degrees of participation in the creation of digital stories impact how people create, share, and learn from digital stories; advances in community-based co-design for rural and under-resourced populations; and explaining how digital stories could impact MCWB within a community. This will enable improved community-based care for MCWB by prioritising learning from health information and experiences beyond access, collaborating and creating community-based content, and mitigating the multiple socio-economic factors impacting mothers in marginalised contexts.

2.1 Research questions

This research will use a co-design approach to develop a novel intervention that leverages DST for the support of MCWB. Experiences and narratives will be elicited to collaboratively achieve this. How the community-based approach is applied, and the preliminary efficacy of the intervention, will be evaluated. The expectation is that the process will generate not only requirements for an intervention appropriate for the context, but also novel community-based approaches for health innovation situated in existing storytelling and cultural practices. The research question is thus:

How can the use of digital mediums to support storytelling for maternal and child well-being through community-based co-design decrease barriers to participation?

This will be addressed by considering the following sub-questions:

1. How are storytelling practices and digital media currently being leveraged and adopted to support MCWB?
 - 1.1. What are the current levels of participation in participatory media in the communities?
 - 1.2. What are ways in which the targeted rural and under-resourced South African communities currently foster storytelling practices, especially on maternal and child well-being?

Lambert [8] describes what he terms a “Taxonomy of Participatory Media Practices” as various levels of participation in creating media that people partake in daily. This ranges from passive media practices, such as surfing the Internet or television channels, to active practices, such as independent creation of digital media and video blogs (vlogs). For the targeted rural and under-resourced communities, these levels of participation will be explored and examined as it relates to storytelling practices, digital and non-digital, that centre around MCWB.

2. How might communities co-design and participate in digital storytelling for maternal and child wellbeing?
 - 2.1 What are prerequisites and barriers to co-design of, and participation in, digital storytelling for people in a rural

and under-resourced South African context, both as consumers and producers?

- 2.2 What are the affordances of digital storytelling towards the support of MCWB?
- 2.3 What are the design requirements of an effective community-based intervention for digital storytelling to support maternal and child wellbeing in a rural and under-resourced context?

3 BACKGROUND

This section serves to explore and extract the relevant information from existing literature applicable to digital storytelling (DST) for maternal and child wellbeing (MCWB) in a community.

3.1 Digital storytelling

Waycott et al. [9] define digital stories as “narratives from digital photographs, video, text, and audio”. This inclusive perspective towards DST caters to storytellers with varying technological and physical abilities, creating a framework in which diverse storytellers and stories are acknowledged. This study considers a broad understanding of what constitutes a digital story, especially as more becomes understood about how stories are told in the study context in rural and under-resourced communities. The creation of local narrative media content has been explored in rural developing contexts including in India [10], Kenya [11] and South Africa [12]. Storytelling and narrative research have been an important part of the decolonial project when researching with indigenous people, countering western traditions of thought and promoting the cultural practices and oral histories of traditionally marginalised groups [13]. Limaye, Rivas-Nieto, Carcamo, and Blas [14] carried out a DST project in rural communities in Peru with community health workers (CHW) and mothers. Using community-based participatory research and community-created content, the researchers were able to reveal community-specific MCH needs and attempt to address them collaboratively. Few other research studies have worked within the MCH and DST domain [15, 16]. There are fewer focusing on how community members create digital stories, and how their current storytelling practices impact the ways different kinds of DST can impact the wellbeing of mothers and children and the community at large.

3.2 Community-based co-design and participatory action research

One of the existing practices within communities, leveraged through community-based participatory action research (CBPAR) and community-based co-design (CBCD), is storytelling. Its inclusive and collaborative nature, involving storyteller, audience, and the further sharing of stories, has been a relatively untapped mode of research in community-based health and wellbeing, especially when focusing on mothers and children. The extensive, well-documented work on indigenous knowledge, narrative practices, and oral histories are evidence that community-based research is valuable and important to the empowerment and development of communities, a key element of DST and CBPAR [17–23]. Digital storytelling has been used often as a generative research activity in CBCD and CBPAR projects, and it has become an emergent

approach to do research with communities [24]. While the DST activities have been carried out to elicit participation and creation of content, not much research has been done to identify and examine how participants experience creating digital stories, how different levels of engagement when facilitating DST workshops affect the created content and experience of participants, and how storytelling through digital media impacts MCWB.

3.3 Maternal and child wellbeing

Maternal and neonatal mortality is considered one of the most significant global health challenges, with low- and middle-income countries (LMICs) contributing alarmingly to the burden [25]. Social media platforms have become increasingly popular for acquiring health information and sharing health stories by mothers [26, 27]. In South Africa, maternal health research is especially pertinent and relevant due to the prevalence of obesity [28], HIV, and diabetes [29] which contributes to adverse maternal outcomes and complications during pregnancy. In rural areas, these maternal outcomes are particularly concerning, in conjunction with the multiple challenges parents face in under-resourced regions. The need for a broader, more holistic, intersectional approach to women's wellbeing has been well documented [30–32], with a focus on women's security, privacy, and safety adding to a perspective on wellbeing. For new and expecting mothers residing in rural contexts, added dimensions to wellbeing exist as they navigate motherhood and caring for themselves and their children in constrained circumstances [4, 15, 33]. When considering women's wellbeing for the design of interventions, Kumar, Karusala, Ismail, and Tuli [6] write that we should “think beyond physiological conditions alone, to consider how to support interrelated freedoms”. Their consideration of intersectional feminism, a lens with which to acknowledge and analyse the complexity of human experience, is particularly intriguing and applicable to exploring innovative and inclusive interventions for under-resourced and rural South African communities. With this in mind, storytelling, more specifically DST, provides an opportunity to explore how peer-created content and information that is context-specific, can support the care of wellbeing of mothers and children holistically, which is where this research is situated.

4 RESEARCH METHODOLOGY

The research methodology for this project will be guided by community-based participatory action research (CBPAR) practices, with intersectional feminist and storytelling modes of research as an analytical lens and foundation, respectively, for research activities. Analysis throughout the research will consist of narrative analysis, specifically thematic and structural analysis. Methods undertaken during this project will largely be visual and arts-based, meaning artistic expressions to communicate human experiences, conforming to storytelling practices.

4.1 Research setting

The study will recruit participants from two communities in South Africa: a rural community in Limpopo, namely Ga-Dikgale and the under-resourced community of Oceanview in Cape Town. Inclusion criteria comprises of mothers, fathers, other caregivers, community health workers, and other relevant stakeholders within the MCWB

ecosystem. Both Ga-Dikgale and Oceanview are communities with: limited access to the Internet [34]; low-income households [34]; and existing relations with the University of Limpopo and the University of Cape Town, respectively. Additionally, the Oceanview community have ambitions to create, disseminate, and own local content [35], and the Ga-Dikgale community embrace a culture of storytelling in their everyday lives, with cultural norms shaping how they engage with new interventions and ideas. For each research site, the following breakdown of participants will be sought across the research activities: 10–12 mothers, 5–10 fathers, 5–10 community health workers, and 5–10 other caregivers and stakeholders. To the best ability of the researchers, provisions will be made to communicate all research activities in the dominant local language of participants by training local community liaisons and researchers to conduct co-design activities, thereafter, translating and interpreting research data for analysis. Having local people lead research activities can work towards the empowerment and knowledge-building of disenfranchised communities [36], which is important in creating an established health content-creating practice within a rural and under-resourced community.

4.2 Structure of research approach

The research activities will be divided into four phases which will be carried out consecutively and influence the activities of each other. These are: *understanding*, *making*, *connecting*, and *embracing*. The purpose of the *understanding* phase is to explore and make sense of the existing storytelling practices within the community as well as to become acquainted with key stakeholders through various engagements. The methods used during this phase are in-depth qualitative interviews and observations. The purpose of the *making* phase is to leverage the existing storytelling practices, illuminated in the *understanding* phase, and explore digital approaches to telling stories. The methods will include diverse, generative, workshops paired with observations of participants during workshops. The *connecting* phase will explore how the community would share and interact with digital stories and a deployment of the digital media according to community preferences. The phase will consist of two one-day workshops concentrating on independent DST and dissemination strategies. Lastly, during the *embracing* phase, the purpose is to assess and understand how the community interacts with digital media in the form of digital stories, and the overall implications it has for DST in rural, under-resourced contexts and for MCWB goals. The activities in this phase will include four follow-up focus group interviews. The activities are summarised in Table 1.

5 PROGRESS TO DATE

Upon the completion of my proposal, I defended it in a presentation on 11 April 2022. The proposal was approved and shortly thereafter I applied for ethics approval from the research ethics council which was subsequently approved in May 2022. I have since embarked on the first phase of my research approach. This included conducting interviews in four rural communities in the Limpopo province during July. Interviews were conducted in the local language, Sepedi, by local intermediaries, as planned. Balancing the need for participants to be comfortable during engagements by speaking

Table 1: Phases of research approach with methods and analysis

Phase	Methods	Analysis
Understanding	Feminist qualitative interviewing Observations	Thematic and structural analysis
Making	Generative DST workshops Observations	Thematic and structural analysis The Transportation Scale (TS-SF) Validity criteria for DST
Connecting	Workshops Observations	Thematic and structural analysis The Transportation Scale (TS-SF) Validity criteria for DST
Embracing	Focus group interviews	Thematic and structural analysis

their preferred language, with my need to be able to dig deeper and engage on topics that are discussed, is something I thinking through as I move towards conducting my first round of interviews with the community of Ocean View, in Cape Town.

5.1 Preliminary findings

During the first engagements with participants, in which I was doing exploratory work into the storytelling landscape, some of the interesting observations included the shared understanding of what storytelling is, or lack thereof. Many of the interviewees would not describe the act of sharing experiences and telling each other stories in conversation as storytelling. When asked about storytelling, they interpreted this means of communication as a more dramatic or artistic form, not one they perform regularly. In this sense they would not describe themselves as storytellers, even though they are doing so in most of their engagements with their community. This is a conflict of semantics, but also of language and personal value. Not seeing oneself as a creator of stories can limit the value one places on oneself as a knowledge producer. This furthers the challenge of creating and sharing stories to recognizing the contribution one can make, and being confident in doing so, merely by relating ones experience with others.

Within the rural communities in Limpopo, interesting potential storytelling challenges arose which will be important to consider as my research continues. This includes the hesitancy to share health information or experiences for fear of gossip and stigmatization. Additionally, experiences besides the ante- and post-natal ones contribute to the overall well-being of parents and their families. These include infertility and associated ostracization or shame, as well as family conflict related to judgment and rejection. These are experiences that, if shared, could lessen the exclusion and loneliness one might feel during the process of wanting a child and parenting a child. More holistic aspects of wellbeing differ across communities, but it is clear that this needs to be understood as I approach the research challenge in my work.

5.2 Other work

In addition to my fieldwork, I have also presented a paper on the importance of accounting for cultural differences in co-design projects, especially as it related to South African communities. This was an

important consideration for my work with diverse communities, and acknowledging the cultural norms that arise both with the application of methods, as well as the responses and participation of the participants. As I embark on the next round of interviews and the workshops, many questions and ideas have already arisen that I need to mull over. This includes investigating who the authors of digital stories are and can be, considering many of the communities who cherish the tradition of storytelling are the elderly with limited access to mobile technology and lack digital literacy. Additionally, the storytelling ecosystem is also not without controversies and limitations, in that gossip is a significant aspect of communal storytelling, which introduces elements of privacy and shame. On the other hand, stories are still considered by many as a primary form of information, health or otherwise, and heavily influence how people act and the decisions they make. Balancing these two components of the storytelling ecosystem is something I need to explore and consider as I further my exploration into this space.

6 CONCLUSION

This document has described the challenges and opportunities of caring for maternal and child wellbeing in an under-resourced context. Through a prioritisation of participants throughout the research activities via community-based participatory co-design, and a focus on narratives expressed through digital media, I believe that the holistic wellbeing of women and children can be cared for better. The exploration of how digital media and stories play a role in the targeted community has the potential of illuminating pathways for innovation that is empathic and sustainable. The expectations are that the research processes will generate not only requirements for an intervention appropriate for the context, but also novel approaches for health innovation and human-computer interaction situated in existing storytelling and cultural practices. I believe that my work and future plans are aligned with the CHI 2023 theme reCHInnecting, since I am exploring ways in which we can work together to support one another using our experiences and expertise on our own lives.

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